Time Estimation

Project: Document Management System (DMS)

Task 2 - Project Planning

Prepared by	Role
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1. Purpose

This document estimates the time needed for the planning activities of the project: A (System Request), B (Feasibility Study), C (Methodology) and D (Time Estimation and Scheduling). The estimates are used to build the PERT chart and the Gantt chart.

2. Estimation Techniques Used

- **Expert judgment:** the team member responsible for the activity gives an estimate based on experience with similar lab work.
- **Bottom-up estimating:** each activity is split into small tasks and their times are added.

3. Activity Breakdown

Activity	Task	Days
A: System Request	A1 Define the business need and requirements	1
	A2 Write the system request and review it with the team	1
B: Feasibility Study	B1 Technical and operational feasibility	2
	B2 Economic analysis (costs, benefits, ROI, BEP, NPV)	2
C: Methodology	C1 Study and compare Agile and Waterfall	2
	C2 Score the options and take the decision	1
	C3 Write the methodology document and review it	2
D: Time Estimation	D1 Estimate durations and set dependencies	1
	D2 Build the PERT and Gantt charts	1
	D3 Review the schedule with the team	1

4. Duration and Effort Estimates

Duration is the number of working days from start to finish. Effort is the total work of all team members in person-days.

Activity	Responsible	Support	Method	Effort (person-days)	Duration (days)
A: System Request	Omar	Michael, Gamal	Expert judgment	3	2
B: Feasibility Study	Omar	Michael, Gamal	Bottom-up	6	4

Activity	Responsible	Support	Method	Effort (person-days)	Duration (days)
C: Methodology	Omar	Gamal, Michael	Bottom-up	7	5
D: Time Estimation	Omar	Gamal	Bottom-up	4	3
Total effort				**20**	

The responsibility follows the task ownership defined in the Staff Plan (Lab 5).

5. Total Project Time for the Planning Phase

Activities A and B can start together because neither needs the other. Activity C waits for B, and activity D waits for both A and C. The longest chain is B, C and D, so the planning phase takes:

Chain	Calculation	Length
A, then D	2 + 3	5 days
B, then C, then D	4 + 5 + 3	12 days

The estimated time of the planning phase is **12 working days**.

6. Time Buffer

Estimates can be wrong, so the team adds a buffer of about 10% (one day) on the critical path. The target completion time for planning is therefore **13 days**.

7. Link with the Whole Project Plan

The 12 days above are the working days of the planning activities only. The whole project follows the 11-week plan from Lab 6, where each lab has one week. The planning activities A, B and C produce the documents of the first labs (system request, methodology and feasibility), and activity D produces the schedule that is later extended into the WBS and timeline of Lab 6.

8. Assumptions

- Team members work part-time on the project, so the effort (20 person-days) is lower than the full capacity of three people over 12 days (36 person-days).
- Reviews are done by a team member who did not write the document.
- No activity is stopped by missing information from the sponsor.